

Wenbo Zhang

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RESEARCH INTERESTS

My research focuses on developing trustworthy large language models (LLMs). I pursue this goal through two complementary directions: (1) **LLM Post-training**: reinforcement learning from human feedback (RLHF) for alignment, reinforcement learning for reasoning, and knowledge distillation; (2) **LLM Evaluation**: effective & efficient evaluation and uncertainty quantification.

EDUCATION

University of California, Irvine

2021.9 - Present

PhD Candidate, Statistics

Donald Bren School of Information and Computer Sciences

Advisor: Prof. Hengrui Cai

University of Washington, Seattle

2019.9 - 2021.3

Master of Science, Biostatistics

Xi'an Jiaotong-Liverpool University

2015.9 - 2019.7

Bachelor of Science, Applied Mathematics

SELECTED PUBLICATIONS AND MANUSCRIPTS

* denotes equal contribution

LLMs & Reinforcement Learning

- [1] [Wenbo Zhang*](#), [Lijinghua Zhang*](#), [Liner Xiang*](#) and [Hengrui Cai](#). “Reasoning Is Not Free: Robust Adaptive Cost-Efficient Router for LLM-as-a-Judge.” [\[ICML,2026\]](#)[\[Code\]](#).
- [2] [Wenbo Zhang](#), [Hengrui Cai](#), and [Wenyu Chen](#). “Beyond the Singular: Revealing the Value of Multiple Generations in Benchmark Evaluation.” [\[ACL Findings, 2026\]](#).
- [3] [Wenbo Zhang*](#), [Zihang Xu*](#), and [Hengrui Cai](#). “Recognizing Limits: Investigating Infeasibility in Large Language Models.” [\[EMNLP Findings, 2025\]](#) [\[Code\]](#).
- [4] [Wenbo Zhang](#) and [Hengrui Cai](#). “Where to Intervene: Action Selection in Deep Reinforcement Learning.” [\[TMLR, 2025\]](#)[\[Code\]](#).
- [5] [Wenbo Zhang](#), [Tong Wu](#), [Yunlong Wang](#), [Yong Cai](#), and [Hengrui Cai](#). “Towards Trustworthy Explanation: On Causal Rationalization.” [\[ICML, 2023\]](#)[\[Code\]](#).
- [6] [Wenbo Zhang](#), [Wenzhuo Zhou](#), [Hengrui Cai](#), and [Zhenglin Qi](#). “Towards Bridging the Gap Between Offline and Iterative LLM Alignment via Preference Distillation.” [\[Under Review\]](#).
- [7] [Lijinghua Zhang*](#), [Liner Xiang*](#), [Wenbo Zhang*](#), [Yiran Zheng](#) and [Hengrui Cai](#). “Large Language Model Alignment with Complex Feedback: A Survey.” [\[Under Review\]](#).
- [8] [Haitao Jiang](#), [Wenbo Zhang](#), [Jiarui Yao](#), [Hengrui Cai](#), [Sheng Wang](#) and [Rui Song](#). “Supervised Fine-tuning versus Reinforcement Learning: A Comprehensive Survey on Large Language Model Post-training.” [\[Under Review\]](#).
- [9] [Wenzhuo Zhou*](#), [Wenbo Zhang*](#), and [Hengrui Cai](#). “Regularized Offline Alignment for Language Models.” Under Review.
- [10] [Jiayi Wang*](#), [Wenbo Zhang*](#), [Wenzhuo Zhou](#), and [Hengrui Cai](#). “PACE: Pessimistic Adaptive Contextual Bandits for Dynamic Optimal Policy Detection.” Under Review.

Statistical Machine Learning

- [11] Eardi Lila, Wenbo Zhang, and Swati Rane. “Interpretable Discriminant Analysis for Functional Data Supported on Random Nonlinear Domains.” [[Journal of the Royal Statistical Society Series B, 2024](#)].
- [12] Lan Shui, Dean Shibata, Kwun Chuen Gary Chan, Wenbo Zhang, Junhyoun Sung, and David R Haynor. “Longitudinal Relationship Between Brain Atrophy Patterns, Cognitive Decline, and Cerebrospinal Fluid Biomarkers in Alzheimer’s Disease Explored by Orthonormal Projective Non-Negative Matrix Factorization.” [[Journal of Alzheimer’s Disease, 2024](#)].
- [13] Youyi Fong, Ying Huang, Bhavesh Borate, Lars Wim and Paul van der Laan, Wenbo Zhang, Lindsay N. Carpp, Iksung Cho, Greg Glenn, Louis Fries, Raphael Gottardo, and Peter B. Gilbert. “Antibody Correlates of Protection From Severe Respiratory Syncytial Virus Disease in a Vaccine Efficacy Trial.” [[Open Forum Infectious Diseases, 2023](#)].
- [14] Lars Van Der Laan, Wenbo Zhang, and Peter Gilbert. “Nonparametric Estimation of the Causal Effect of a Stochastic Threshold-based Intervention.” [[Biometrics, 2022](#)].
- [15] Sifan Song, Tianming Bai, Yanxin Zhao, Wenbo Zhang, Chunxiao Yang, Jia Meng, Fei Ma, and Jionglong Su. “A New Convolutional Neural Network Architecture for Automatic Segmentation of Overlapping Human Chromosomes.” [[Neural Processing Letters, 2021](#)].
- [16] Wenbo Zhang, Kwun Chuen Gary Chan, Dean Shibata, and David Haynor. “Finding Atrophy Patterns of Grey Matter Through Orthonormal Non-negative Factorization.” [[Medical Imaging: Image Processing, 2021](#)].
- [17] Peter Gilbert, David Montefiori, Adrian McDermott, Youyi Fong, . . . , and United States Government (USG)/CoVPN Biostatistics Team (Wenbo Zhang). “Immune Correlates Analysis of the mRNA-1273 Covid-19 Vaccine Efficacy Clinical Trial.” [[Science, 2021](#)].

Industry Experience

Amazon, Applied Scientist Intern

Summer 2025

Mentored by Prof. Sheng Wang, Prof. Rui Song and Dr. Cynthia Lu

Topic: Reward modeling and Post-training for LLM Agent.

- Develop a two-stage selective framework that leverages list-wise feedback to improve reward model learning.
- Apply the method in production for rationale judgment and propose RL methods to improve LLM agent.

Meta, Central Applied Science Team, Research Scientist Intern

Summer 2024

Mentored by Dr. Wenyu Chen

Topic: Difficulty Quantification for Large Language Models (LLM) Benchmark.

- Developed difficulty metrics (P-correct) as a fine-grained difficulty score for individual prompts.
- Designed LLM-tagging-based difficulty metrics and analyzed a wide range of open-source benchmark datasets.

IQVIA, Advanced Analytics, Machine Learning Research Intern

Summer 2022

Mentored by Dr. Tong Wu and Dr. Yunlong Wang

Topic: Interpretable Neural Sequence Prediction Models

- Developed a selective rationalization approach for language models (BERT) to explain the predictions by leveraging two causal desiderata, non-spuriousness, and efficiency.
- Applied the method to real-world text and Electronic Health Records (EHR) datasets.

Research Experience

RACER: Robust Adaptive Cost-Efficient Router

2025.3-present

Department of Statistics, University of California, Irvine

Advised by Prof. Hengrui Cai

- Investigated whether reasoning skills learned from problem-solving tasks effectively transfer to judgment tasks.
- Developed a tractable method that jointly addresses the cost-performance trade-off and robustness to distribution shift via constrained distributionally robust optimization.

RLHF for Large Language Model Alignment

2024.2-2025.5

Department of Statistics, University of California, Irvine

Advised by Prof. Wenzhuo Zhou and Prof. Hengrui Cai

- Designed controlled experiments to understand the performance gap between offline and iterative direct preference optimization (DPO) algorithms.
- Designed an offline optimization method that distills knowledge from an explicit preference model to the policy model, achieving alignment without the computational overhead of iterative methods.
- Implemented a max-minimax algorithm to utilize importance ratios and pessimism to mitigate overoptimization in offline alignment.

Uncertainty Quantification for Large Language Models

2023.9-2024.10

Department of Statistics, University of California, Irvine

Advised by Prof. Hengrui Cai

- Developed an infeasible benchmark to assess LLMs' refusal capabilities and self-confidence.
- Fine-tuned models to enhance their refusal ability in terms of infeasible tasks.

Reinforcement Learning with High-Dimensional Action Space

2023.1-2024.9

Department of Statistics, University of California, Irvine

Advised by Prof. Hengrui Cai

- Designed a variable selection method based on conformal inference to find the sufficient and necessary action set from offline trajectory data.
- Developed a hard mask strategy to incorporate offline variable selection results with deep policy/value networks to make online learning more efficient with less spurious features.

Fellowships & Awards

School of Public Health's Outstanding MS Student Award, awarded to one master student in the Department of Biostatistics every year, *University of Washington*.

UW Summer Institutes Scholarship, *University of Washington*.

University Academic Achievement Award, awarded to 10% of all undergraduates, *XJTLU*.

Talk

Responsible NLP group at Meta

2024.12

Beyond the Singular: the Essential Role of Multiple Generations in Benchmark Evaluation

UCI CausalRL+X Reading Group

2024.4

Mitigating Reward Overfitting in RLHF

The Fortieth International Conference on Machine Learning

2023.7

Towards Trustworthy Explanation: On Causal Rationalization.

The Thirty-Sixth Annual Conference on Neural Information Processing Systems (Causal Machine Learning for Real-World Impact Workshop)

2022.12

Towards Trustworthy Explanation: On Causal Rationalization.

Profession Activity

Conference Reviewer:

ICLR (2025-2026), AISTATS (2025), ICML (2024-2026), NeurIPS (2024-2025)

Journal Reviewer:

Journal of the American Statistical Association ($\times 1$), Statistical Analysis and Data Mining ($\times 1$), Journal of Applied Statistics($\times 1$)

Skills

- **Programming:** Python, PyTorch, R, SQL, Linux, Matlab
- **Tools:** Git, AWS